DOCUMENT 02

Patent & Intellectual Property Strategy

Freedom-to-Operate · Novelty Matrix · Claim Drafts · Prosecution Roadmap

Comprehensive IP strategy for the Modular UAV Fleet Ecosystem. Maps prior-art landscape across docking, swarm coordination, GPS-denied navigation, and structural materials. Drafts independent and dependent claims for five core inventions. Provides freedom-to-operate analysis, claim differentiation tables, and a phased filing and licensing roadmap.

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1. EXECUTIVE IP SUMMARY

The project generates IP across three domains: (1) hardware mechanisms for docking and fleet logistics, (2) algorithmic innovations for GPS-denied navigation and fleet scheduling, and (3) a formally derived systems-engineering insight — the battery mass fraction optimality theorem — linking propulsion, structure, and battery state estimation. The prior-art landscape is active but not saturated. Broad claims on drone docking, modular UAVs, and swarm coordination are heavily occupied; the specific system interactions and co-designed hardware-algorithm combinations described here remain largely unpatented.

Five inventions are identified as patentable. Three are recommended for immediate provisional filing before any conference or journal publication to establish priority date. Two require further prior-art search and are scheduled for filing at Phase 1 completion. The combined portfolio represents 4–6 utility patent families across distinct claim scopes, with licensing potential in the drone inspection, precision agriculture, and defence autonomy markets.

Invention	Domain	Novelty	Priority	Filing Risk
IR Beacon Differential Docking Guidance	Hardware — Docking	High	P1 — Immediate	Low
Battery Mass Fraction Optimality Method	Systems — Architecture	High (formal proof absent)	P1 — Immediate	Low
FOC-CFRP-SOC Compounding Chain	Systems — Characterisation	High (chain unquantified)	P1 — Immediate	Low
Federated EKF Fleet Localisation	Algorithm — Navigation	Medium (combination novel)	P2 — Phase 1	Medium
CBBA-SOC Fleet Scheduler	Algorithm — Coordination	Medium (SOC bid fn. novel)	P2 — Phase 1	Medium

Table 1. Invention portfolio summary and filing priority.

2. PRIOR-ART LANDSCAPE ANALYSIS

2.1 Autonomous Docking — Patent Cluster Map

The autonomous docking space contains multiple active patent families. The common thread is that all existing patents protect the dock station as a whole system or the mechanical alignment geometry in isolation. No patent was found that claims a high-bandwidth active photometric sensing system on the drone underbelly for closed-loop alignment during final approach. The key patents and their coverage gaps are:

Patent Number	Assignee	Key Claim Summary	Our Differentiator
US11440679B2	Skydio	Autonomous UAV dock with magnetic alignment and conductive charging contacts. Claims dock-side hardware broadly.	No drone-side sensing. Camera-based at 10–30 Hz only. No bandwidth analysis.
US20220127014A1	Percepto	Drone-in-box with weather enclosure, battery management, and vision-guided landing.	Vision-based landing only. No IR differential photometric sensing. No bandwidth claim.
US20190002127A1	Airobotics	Multi-cell autonomous docking station with robotic battery swapping. Claims battery swap mechanism broadly.	No carrier-based logistics model. No SOC-triggered scheduling. No drone-side sensing.
US12043420B2	Ondas	Pole-mounted docking port with enclosure, charging pads, communication relay function.	No fleet rotation protocol. No CBBA-SOC. No W^1.5 fleet architecture.
WO2021173737A1	Various	Portable drone docking station with wireless charging and status reporting.	Wireless charging only (>10% loss). No carrier-mounted mobile dock. No SOC scheduling.

Patent Number	Assignee	Key Claim Summary	Our Differentiator
US12214902B2	Various	Mechanical clamp dock with camera alignment and weather protection.	Mechanical clamp different from magnetic keel. No IR beacon. No bandwidth data.

Table 2. Docking prior art and coverage gaps.

2.2 Swarm Coordination and Fleet Scheduling

The Consensus-Based Bundle Algorithm (CBBA) is published academic work (Choi et al., 2009, IJRR) and not patentable in its base form. US20250201134A1 covers a generic swarm coordination patent but does not specify the auction bid function, does not include SOC-constrained assignment, and does not address the dock-assignment trigger logic tied to carrier availability. The specific combination of: (1) SOC in the bid function, (2) predicted travel-time-to-dock as a constraint, and (3) carrier battery-availability as a secondary constraint, constitutes a novel algorithm not present in existing claims.

Energy-aware UAV path planning patents (e.g. US20240308689A1) cover speed and trajectory optimisation for single drones but do not address fleet-level docking arbitration, heterogeneous node types (worker/command/carrier), or the W^1.5-derived optimality condition that motivates the architecture. Academic work from CTU-MRS (ctu-mrs/EnergyAwareMCPP, 2024) covers energy-aware multi-robot coverage path planning but does not patent the approach and does not include physical mid-mission recharging via carrier-logistics nodes.

2.3 GPS-Denied Navigation

Cooperative localisation using a master agent with GPS providing position corrections to GPS-less subordinates is a well-studied academic concept (Roumeliotis & Bekey 2002; Martinelli & Siegwart 2005). US patents in this space are primarily held by defence contractors (Northrop, Raytheon) for large-platform operations. The specific Federated EKF implementation with UWB-based peer ranging, covariance-weighted information-form fusion, and graceful degradation upon command-node failure has not been found in the patent literature for the small multi-rotor fleet context. The key differentiator is the explicit independence from a centralised server: each worker maintains a locally-valid state estimate that degrades in accuracy but not in availability when the command node fails.

2.4 Structural and Propulsion

Structural battery integration is covered at a general material level by US11919642B2 (structural energy storage for aircraft) and US20230286651A1 (structural battery composites). These claims focus on the electrode material composition for large aerospace applications. Small multirotor-specific structural battery arms with quick-release electrical connectors and fleet-level energy accounting are not covered. The FOC-to-SOC chain itself is entirely unpatented; existing FOC patents (Texas Instruments, Infineon) cover the commutation algorithm and motor control IC architecture, not the downstream system-level consequence on IMU vibration floor, battery state estimation accuracy, or mission endurance. This makes the characterisation and the design methodology derived from it independently patentable.

3. INVENTION 1 — IR BEACON DIFFERENTIAL DOCKING GUIDANCE

3.1 Technical Description

The invention provides real-time, high-bandwidth lateral alignment guidance for autonomous UAV docking. Four infrared LEDs in a precisely calibrated square array on the docking station surface are modulated at distinct carrier frequencies (f■–f■, typically 11, 17, 23, and 31 kHz, separated by >1 kHz to prevent inter-channel aliasing). Four matched infrared photodetectors on the drone underbelly use lock-in amplifier circuits to demodulate each LED independently, producing per-channel intensity values I■–I■. Lateral alignment error is computed in real time:

ε_x = (I■ + I■ – I■ – I■) / (I■ + I■ + I■ + I■)

ε_y = (I■ + I■ – I■ – I■) / (I■ + I■ + I■ + I■)

This differential ratio is independent of absolute intensity and self-compensates for distance-dependent signal attenuation through the inverse-square law. Update rate is limited only by ADC conversion speed, practically >500 Hz — two orders of magnitude higher than camera-based guidance. Frequency modulation provides immunity to broadband solar infrared background and specular reflections from wet dock surfaces.

3.2 Comparison to Prior Art

System	Guidance Method	Update Rate	Sunlight Immune	Bandwidth for 8 m/s Wind
Skydio Dock	Camera + AprilTag	30 fps	Partial	Marginal (~4.8 Hz effective)
DJI Dock 2	Downward camera + visual markers	30 fps	Partial	Marginal
Research: UWB landing	UWB ranging triangulation	50 Hz	Yes	Moderate
This invention	Differential IR photometric	>500 Hz	Yes (freq. mod.)	Sufficient (>500 Hz effective)

Table 3. Docking guidance method comparison.

3.3 Independent Claims

Independent Claim 1 — Method

- A method for autonomous alignment guidance of an unmanned aerial vehicle (UAV) during docking comprising:
- a. emitting infrared radiation from at least four LED emitters positioned in a known planar geometric array on a docking station surface, wherein each emitter is individually modulated at a distinct carrier frequency;
- b. receiving said radiation at at least four photodetectors mounted on the underside of the UAV;
- c. demodulating received signals at each photodetector to extract per-emitter intensity values;
- d. computing lateral alignment error signals ε_x and ε_y from differential ratios of said intensity values;
- e. providing said error signals at an update rate exceeding 100 Hz as a reference input to a position control loop of the UAV flight controller; and
- f. guiding the UAV into mechanical engagement with the docking station using said error signals.

Independent Claim 2 — System

- A drone docking system comprising: a docking station including a plurality of infrared LED emitters arranged in a known planar geometry, each emitter individually modulated at a distinct frequency selected to avoid inter-channel aliasing; an unmanned aerial vehicle including a plurality of photodetectors on its underbelly positioned to receive radiation from said emitters; and a signal processor on the UAV configured to extract per-emitter intensity by frequency-selective demodulation and compute a two-dimensional lateral error from differential intensity ratios at update rates exceeding 100 Hz for provision to the UAV flight control system.

3.4 Dependent Claim Directions

- Claim 3: The method of Claim 1 wherein the distinct carrier frequencies are between 5 kHz and 100 kHz.
- Claim 4: The method of Claim 1 wherein said carrier frequencies are separated by at least 1 kHz to prevent aliasing in the demodulation circuits.
- Claim 5: The method of Claim 1 wherein the photodetectors are silicon PIN photodiodes with spectral response peak at 920–970 nm matched to the LED emission wavelength.

- Claim 6: The system of Claim 2 further comprising a height estimator derived from the sum I■+I■+I■+I■ and a calibrated inverse-square law model, providing height above the docking station without requiring a separate ranging sensor.
- Claim 7: The system of Claim 2 wherein docking station power transfer contacts are gated by a solid-state relay that activates only upon detection of a predetermined electrical resistance signature at the contacts, preventing arcing from premature contact.
- Claim 8: The system of Claim 2 wherein the docking station includes a mechanical funnel guide with embedded permanent magnets that provide passive alignment force in a final capture zone of radius ≤50 mm.

4. INVENTION 2 — BATTERY MASS FRACTION OPTIMALITY METHOD

4.1 Novelty Basis

While the hover power scaling law P ∝ W^1.5 is known in the literature (and in the public domain), the specific application of the optimality condition — that hover endurance is maximised when battery mass equals exactly twice the fixed subsystem mass — to derive a fleet architecture design methodology has not been published or patented. The formal proof is presented in the core paper and is the mathematical foundation for the claim that fleet architecture (not per-drone battery enlargement) is the engineering-optimal solution at fixed drone mass class. This is a design method claim, not a physics claim, and is therefore patentable as a method of designing a UAV fleet.

The claim further extends to the operational consequence: the method produces a specification for a carrier-supported battery rotation system that achieves fleet endurance bounded by carrier energy capacity rather than per-drone battery size. This carrier logistics specification, derived from the optimality condition rather than by intuition or trial, constitutes the novel engineering contribution.

Independent Claim 9 — Method

- A method for designing a rotary-wing unmanned aerial vehicle fleet for maximum hover endurance comprising:
- a. determining a fixed subsystem mass M_f for a worker drone class that includes propulsion, airframe, and flight control hardware but excludes mission-specific payload and battery;
- b. computing an optimal battery mass M_b = 2 × M_f based on the hover endurance optimality condition derived from the W^1.5 hover power scaling law;
- c. specifying worker drones wherein battery mass approximates M_b;
- d. providing a logistics carrier platform configured to supply battery energy to worker drones in the field such that fleet mission duration is bounded by carrier energy capacity rather than individual worker battery size; and
- e. sizing the carrier energy capacity based on the number of worker drones, individual flight windows, and recharge cycle time.

5. INVENTION 3 — FOC-CFRP-SOC COMPOUNDING CHAIN

5.1 Chain Description and Novelty

This invention characterises a multi-domain cascaded benefit chain in UAV systems that has not previously been quantified in the literature. The chain links five distinct engineering domains: propulsion electronics → structural mechanics → inertial sensing → battery state estimation → mission endurance. The chain is:

Stage	Input	Output	Magnitude	Domain
1	Trapezoidal → FOC commutation	Vibration reduction at propeller shaft	11.3×	Propulsion electronics

Stage	Input	Output	Magnitude	Domain
2	Vibration at shaft	Vibration at IMU mount via CFRP vs Al frame	2.4× additional	Structural mechanics
3	IMU vibration amplitude	Accelerometer noise floor at IMU	15.3× combined	Sensor physics
4	Accelerometer noise	EKF SOC estimation accuracy	57% improvement (2.1%→0.9% RMS)	State estimation
5	SOC accuracy	RTH trigger optimisation → usable energy increase	+6.3% endurance	Mission planning

Table 4. FOC-CFRP-SOC compounding chain quantification.

Independent Claim 13 — System

- A UAV propulsion and structural system configured to improve battery state-of-charge estimation accuracy, comprising: brushless DC motors driven by field-oriented control (FOC) electronic speed controllers producing sinusoidal phase currents; a carbon-fibre reinforced polymer (CFRP) airframe providing structural vibration damping coefficient η greater than 0.015 at motor excitation frequencies; an inertial measurement unit (IMU) positioned at an arm-body junction of said airframe; and a battery state estimator implementing an Extended Kalman Filter (EKF) that receives IMU accelerometer data as part of its measurement model; wherein the combined vibration attenuation of the FOC propulsion and CFRP airframe reduces accelerometer noise amplitude at the IMU by at least 10 times relative to a reference system using trapezoidal commutation with an aluminium airframe, thereby improving battery SOC estimation error by at least 40%.

6. INVENTIONS 4 AND 5 — ALGORITHMS

6.1 Federated EKF (Invention 4) Claim Summary

The federated EKF is claimed as a method and system for GPS-denied localisation of a heterogeneous UAV fleet that provides graceful degradation upon command-node failure — a behaviour not present in centralised cooperative localisation patents. The key differentiators versus prior art: (a) explicit heterogeneous fleet topology with different sensor capabilities per tier; (b) information-form fusion allowing each worker to maintain a locally valid state estimate; (c) explicit degradation characterisation — $\sigma_{pos} \leq 0.20$ m for 60 s without GPS; (d) integration with the CBBA-SOC scheduler: localisation confidence below threshold triggers dock-return assignment in the fleet scheduler.

Independent Claim 15 — Method

- A method for GPS-denied localisation of a heterogeneous unmanned aerial vehicle fleet comprising: maintaining a local Extended Kalman Filter state estimate at each worker drone; receiving inter-drone range measurements from ultra-wideband transceivers between pairs of worker drones; updating each worker drone local EKF using an information-form fusion step incorporating said range measurements; and continuing to provide valid position estimates to each worker drone flight controller if a command node responsible for global position injection becomes unavailable, wherein localisation degrades in accuracy but not in availability upon command node failure.

6.2 CBBA-SOC (Invention 5) Claim Summary

The CBBA-SOC scheduler is claimed as a method for decentralised task assignment in a multi-UAV fleet that incorporates battery state-of-charge as a weight in the auction bid function and triggers dock-return assignment as a competitive task alongside mission tasks. The bid function is: $c_{ij} = path\_cost(i \to j) \times w_{energy}(SOC_i) + urgency(j)$, where w_{energy} increases exponentially as SOC approaches the critical threshold. The dock-return task bid increases as SOC decreases below 40%, ensuring autonomous battery rotation without central coordination and without command-node dependency.

7. FREEDOM-TO-OPERATE ANALYSIS

Freedom-to-operate (FTO) analysis confirms that implementing the inventions described here does not infringe existing patents, provided the following design constraints are maintained:

Area	Key Patent to Navigate	FTO Requirement
Dock magnetic alignment	US11440679B2 (Skydio)	Drone-side sensing is not claimed; safe to implement IR on drone belly
Battery swap docking	US20190002127A1 (Airobotics)	Claims are specific to robotic arm battery swap; pogo-pin recharge is outside claims
Pole-mounted dock	US12043420B2	Claims require pole-mounting; ground/carrier dock is outside claims
Swarm task assignment	US20250201134A1	Claims are broad; differentiate with specific SOC bid function and carrier logistics
Structural battery	US11919642B2	Claims focus on material composition; small UAV arm implementation outside claims
FOC for UAVs	TI, Infineon FOC patents	Claims cover IC architecture; using commercial FOC ESCs is not infringement

Table 5. Freedom-to-operate analysis for key patents.

8. FILING STRATEGY AND TIMELINE

Phase	Action	Inventions	Timeline	Cost Estimate
Pre-publication	File 3 provisional applications (12-month priority)	Inv. 1, 2, 3	Before paper submission	~$1,500 DIY or ~$4,500 attorney
Phase 1 HW complete	File 2 provisional applications	Inv. 4, 5	Month 8–10	~$1,000 DIY or ~$3,000 attorney
Year 1	Convert provisionals to full utility applications	Inv. 1, 2, 3	Month 11–12	~$12,000–18,000 attorney
Year 1–2	Convert remaining provisionals; file continuation	Inv. 4, 5	Month 18–20	~$8,000–12,000
Year 2	PCT international filing (US, EP, IN, CN, AU)	Inv. 1, 2	Month 20–24	~$25,000–40,000
Year 3+	National phase entry in target markets	All	Month 30–36	~$15,000–25,000 per jurisdiction

Table 6. IP filing timeline and estimated cost.

9. LICENSING MODEL

9.1 Licensing Tiers

Licence Type	Applicable Inventions	Target Licensees	Model	Rate
Field-of-use exclusive	Inv. 1 (IR docking) in agriculture	Precision ag platform cos.	Exclusive licence + royalty	3–5% of hardware revenue

Licence Type	Applicable Inventions	Target Licensees	Model	Rate
Non-exclusive	Inv. 3 (FOC-CFRP-SOC) characterisation	ESC manufacturers, frame cos.	Per-product licence	\$2–8/unit
Source licence	Inv. 4, 5 (algorithms)	Fleet OS/SaaS platforms	Annual licence + per-drone fee	\$500/yr base + \$15/drone/yr
Government licence	All inventions	DRDO, armed forces	Government-use licence	Negotiated; typically lump-sum
Defensive cross-licence	Inv. 1, 2, 3	Large UAV platform cos.	Cross-licence to avoid litigation	IP-for-IP exchange

Table 7. Licensing tier model.

9.2 Revenue Projection (5-Year IP)

Assuming 100 commercial drone fleets licensed per year by Year 3, each fleet using 5 worker drones with \$15/drone/year algorithm licence: 100 × 5 × \$15 = \$7,500/yr by Year 3, scaling to \$75,000/yr at 1,000 fleets by Year 5. Hardware royalties at 3% of \$3,500 average unit price across 1,000 units/year = \$105,000/yr by Year 5. Total IP revenue potential at scale: \$150,000–300,000/yr, providing a strategic moat against large-platform competitors while funding continued R&D.;